NCI
Coláiste Náisiúnta na hÉireann | National College of Ireland

This certificate is awarded to:

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FOR COMPLETING THE PROGRAMME

Academic Integrity: For Students, Second Edition

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epigeum

Grok

Data Structures and Algorithms CA1 Report

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GitHub repo: https://github.com/NowakOlivier5/DSA-CA1

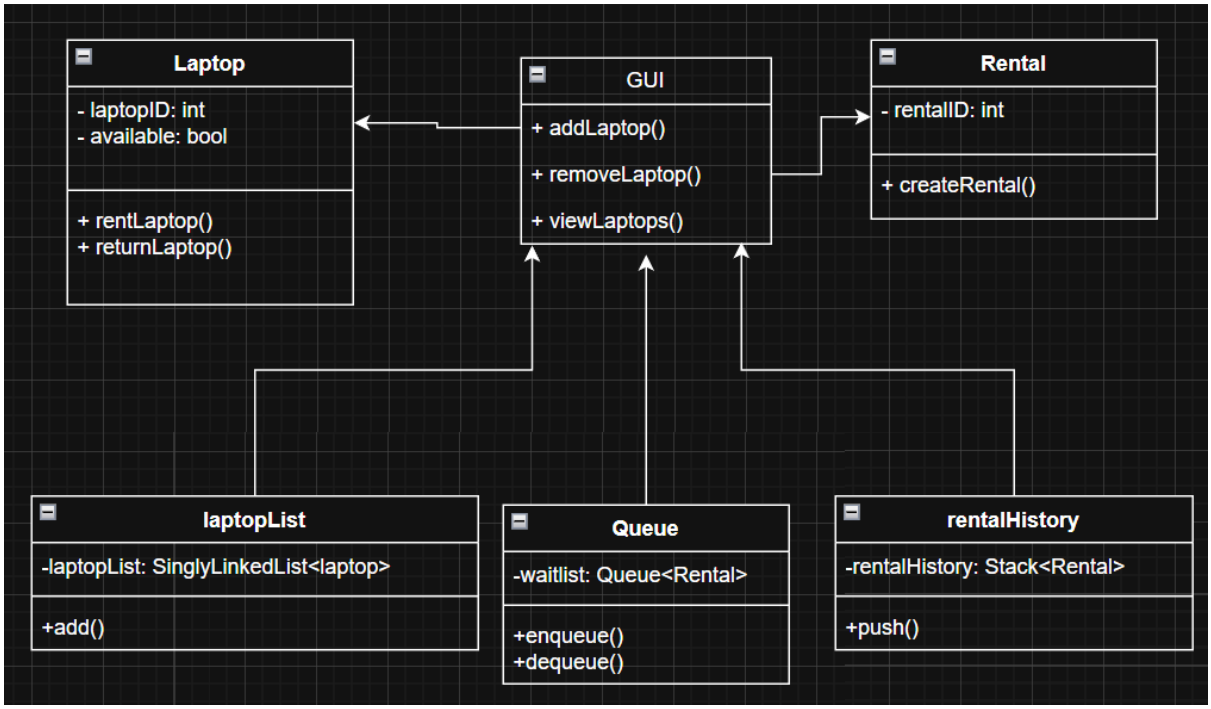
I chose the Digital Tools/Tech Access them for my project, as it's the main theme I could think of ideas for. The problem I am focusing on is the people who don't have access to modern technology. I decided on making a library style laptop rental system, similar to a surface tablet rental scheme in my primary school when I was a kid. Having access to these when I was younger helped me learn how to use windows and various common programs such as Word and PowerPoint, which I otherwise would not have been able to learn as we did not have a computer at home. I would also attribute this early learning to me developing an interest in computers and being where I am now and believe other children should also have this opportunity to learn and hopefully discover an interest in technology like I had back then.

The app will serve as a way to give access to laptops to primary school children, to help with learning and give access to the internet. Various skills gained from this can help in future careers and general life, as technology is being integrated everywhere in our daily lives and workplaces.

The features will allow you to borrow a laptop, return a laptop, see available laptops, view loan history, and add a laptop to the list. If there are no laptops left you are added to a waiting list.

The apps gui will consist of: View available laptops, borrow a laptop, return a laptop, view loan history, add a laptop to the list, and view position in queue.

The software will be going off the assumption that it's a book return machine like those found in libraries, but instead of scanning the barcode you type the code manually.


Laptop
- laptopID: int
- available: bool
+ rentLaptop()
+ returnLaptop()
GUI
+ addLaptop()
+ removeLaptop()
+ viewLaptops()
Rental
- rentalID: int
+ createRental()
laptopList
-laptopList: SinglyLinkedList<laptop>
+add()
Queue
-waitlist: Queue<Rental>
+enqueue()
+dequeue()
rentalHistory
-rentalHistory: Stack<Rental>
+push()